We can maximize each factor separately. The left factor has maximum $\frac{1}{2}$ while the right has maximum $\sqrt{5}$ (Cauchy).

Hence our maximum is $\frac{\sqrt{5}}{2}$.

31. If $120^4 + 272^4 + 315^4 + n^4 = 353^4$ then find 'n' ?

Sol:

$120^4 + 315^4 + n^4$

$= (353^2 + 272^2)(353 + 272)(353 - 272) = (353)(15)^4$

$8^4 + 21^4 + \left(\frac{n}{15}\right)^4 = 353^2 + 272^2$

$64^2 + 441^2 + \left(\frac{n}{15}\right)^4 = 353^2 + 272^2$

$\left(\frac{n}{15}\right)^4 + 441^2 - 353^2 = 272^2 - 64^2$

$\left(\frac{n}{15}\right)^4 = (208)(336) - (794)(88)$

$\left(\frac{n}{15}\right)^4 = 16[(52)(84) - (397)(11)] = 1 \Rightarrow$ **n = 30.**

32. If reals a, b, c satisfy $a^2 + b^2 + c^2 = 3$ then prove that $|a| + |b| + |c| - abc \le 4$?

Sol :

By AM-GM we have

$(|a| + |b| + |c|)^2 \le 3(a^2 + b^2 + c^2) = 9$

$\Rightarrow |a| + |b| + |c| \le 3$ and $a^2 + b^2 + c^2 \ge \sqrt[3]{a^2b^2c^2}$

so $a^2 b^2 c^2 \le 1 \Rightarrow |abc| \le 1$

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with $0 \le x, y, z, w \le 1$

(draw a diagram and this will be more obvious; it's just the Pythagorean Theorem).

Then $\sum a^2 = \sum (x^2 + (1-w)^2) = 2\left(\sum (x - \frac{1}{2})^2\right) + 2$

$0 \le x \le 1 \Rightarrow 0 \le (x - \frac{1}{2})^2 \le \frac{1}{4} \Rightarrow 2 \le \sum a^2 \le 4.$

12. Suppose there are l, m, n $\in \mathbb{R}$ satisfying the equations m - n = 8 ; mn + l^2 + 16 = 0 . Find $m^{2007} + n^{2007} + l^{2007}$

Sol:

We have m = n + 8 so substituting this **mn + l^2 +16 = 0** we get $l^2 + (n + 8)n + 16 = 0$ or $l^2 + (n + 4)^2 = 0$ consequently $l = 0$ and n = - 4

From the relation m = n + 8 we find easily m = 4

The value of $m^{2007} + n^{2007} + l^{2007}$ is obviously zero.

13. Let A = 444...44 (2n digits) and B = 888...88 (n digits). Show that A + 2B + 4 is a perfect square.

Sol :

$$A = \frac{4(10^{2n} - 1)}{9}, \quad B = \frac{8.10^n - 8}{9}$$

$$A + 2B + 4 = \frac{4(10^{2n} - 1) + 16.10^n - 16 + 36}{9} = \frac{[2(10^{2n} + 2)]^2}{9}$$

14. Solve $\frac{\text{Sin}x}{\text{Sin}y} = \sqrt{2}$ **and** $\frac{\text{Tan}x}{\text{Tan}y} = \sqrt{3}$

Sol :

$$\frac{\text{Tan}x}{\text{Tan}y} = \frac{\text{Sin}x}{\text{Sin}y} \cdot \frac{\text{Cos}y}{\text{Cos}x} = \sqrt{2} \cdot \frac{\text{Cos}y}{\text{Cos}x} = \sqrt{3}$$

so that $1 + |x| - 4|x|^2 \le \frac{17}{16} \Rightarrow |4x^2y - y - x| \le \frac{17}{16}$.

10. A certain triangle has angles (180-3x) ,2x , x and opposite Sides respectively. Find its perimeter.

Sol :-

By the law of sines, we obtain

$$\frac{n+1}{\sin 3x} = \frac{n+2}{\sin 2x} = \frac{n}{\sin x} = \frac{1}{\sin 2x - \sin 3x} = \frac{1}{\sin 3x - \sin x}$$

$\Rightarrow$ Sin2x - Sin3x = Sin3x -Sinx $\Rightarrow$ Sin2x + Sinx = 2Sin3x

After simplification we get $\text{Cos}^2\left(\frac{x}{2}\right) = \frac{7}{8}$

Sinx = $\frac{\sqrt{7}}{4}$, Sin2x = $\frac{6\sqrt{7}}{16}$,Sin3x = $\frac{5\sqrt{7}}{16}$

So we have n = 4 and the perimeter is 15

11. Find $ax^5 + by^5$ if the real numbers a, b, x, y, satisfy the equations $ax + by = 3$, $ax^2 + by^2 = 7$,$ax^3 + by^3 = 16$, $ax^4 + by^4 = 42$

Sol : -

Let S = x + y and P = xy.We have

$(ax^2 + by^2)(x + y) = ax^3 + by^3 + xy\,(ax + by)$

$7S = 16 + 3P$ $(ax^3 + by^3)(x + y) = ax^4 + by^4 + xy(ax^2 + by^2)$

16S = 42 + 7P. Solving this system for S and P, we obtain

S = - 14 and P = -38.

We have $(ax4 + by^4)(x + y) = ax^5 + by^5 + xy(ax^3 + by^3)$

$\Rightarrow -42.14 = ax^5 + by^5 - 38.16$

$ax^5 + by^5 = 38.16 - 42.14 = 20.$

9. Find four digit number *abcd* such that 4(*abcd*) = *dcba*.

Sol: If $a \geq 3$ then 4(*abcd*) be comes five digit number.
Hence $a < 3$ ------(i)
4(*abcd*) = *dcba* so it is even
$\Rightarrow$ *a* is even and less than 3 from (i)
hence $a = 2$
So, the equation concludes to 4(2 *b c d*) = *d c b* 2. -----(ii)
$\Rightarrow d \geq 8 \Rightarrow d = 8$ or 9
If take $d = 9$ then units place of R.H.S should be 6.
But from eq - (ii) units place is 2
therefore $d \neq 9 \Rightarrow d = 8$.
Now equation reduces to 4(2 *b c* 8) = 8 *cb* 2.

$\Rightarrow \quad 4(2000+100b+10c+8) = 8000+100c+10b+2$

$\Rightarrow \quad 8000+400b+40c+32 = 8000+100c+10b+2$

$\Rightarrow \quad 400b+40c+30 = 100c+10b$

$\Rightarrow \quad 40b+4c+3 = 10c+b$

$\Rightarrow \quad 39b-6c+3 = 0$

$\Rightarrow \quad 13b+1 = 2c$

$c \leq 9 \Rightarrow 2c \leq 18 \Rightarrow 13b+1 \leq 18$

$\Rightarrow b = 1 \Rightarrow 2c = 14 \Rightarrow c = 7$

$\therefore a\,b\,c\,d = 2178.$

10. Determine the integer *n* for which $n^2+19n+92$ is a square.

Sol: Let $n^2+19n+92 = x^2$ where *x* is a positive integer.

Now $4(n^2+19n+92) = 4x^2$

$\Rightarrow (2n+19)^2+7 = 4x^2 \Rightarrow \quad (2x)^2-(2n+19)^2 = 7$

or $(2n+19)^2 = 4x^2-7$

That is $4x^2-7$ must be the square of an integer.

Let $4x^2-7 = k^2$ for some integer *k*.

$4x^2-k^2 = 7 \Rightarrow (2x-k)(2x+k) = 7$

Since 7 is an integer, so

37. Solve the system of equations $x + y + z = 6$, $xy + yz - zx = 7$, $x^2 + y^2 + z^2 = 14$.

Sol:

$$xy + yz + zx = \frac{(x+y+z)^2 - (x^2+y^2+z^2)}{2} = 11$$

From $xy + yz + zx = 11$ and $xy + yz - zx = 7$

we get $xy + yz = 9$ and $xz = 2$.

From $xz = 2$ and $x + y + z = 6$ we get

Further we get $y^2 + xy + yz - 6y = 0$

$y^2 + 9 - 6y = 0 \Rightarrow (y - 3)^2 = 0$ so $y = 3$.

then $x + z = 3$ and so all $(x, y, z) = (1, 3, 2)$ and $(2, 3, 1)$

38. Given that

$$\mathbf{x_1 + x_2 + x_3 = y_1 + y_2 + y_3 = x_1y_1 + x_2y_2 + x_3y_3 = 0}$$

Prove that $\dfrac{x_1^2}{x_1^2+x_2^2+x_3^2} + \dfrac{y_1^2}{y_1^2+y_2^2+y_3^2} = \dfrac{2}{3}$

Sol:

$x_1 + x_2 + x_3 = 0$....(1) , $x_1y_1 + x_2y_2 + x_3y_3 = 0$.....(2)

$x_1 + x_2 + x_3 = 0 \Rightarrow x_2 + x_3 = -x_1$

Substituting (1) in (2) we get

$(x_1, x_2, x_3) = k(y_3 - y_2, y_1 - y_3, y_2 - y_1)$ for some k

So the expression 'simplifies' to

$$\Rightarrow \frac{(y_3-y_2)^2}{2\left(y_1^2+y_2^2+y_3^2-y_1y_2-y_1y_3-y_2y_3\right)} + \frac{y_1^2}{y_1^2+y_2^2+y_3^2}$$

And because $y_1 + y_2 + y_3 = 0$, this equals:

$$\frac{(y_3-y_2)^2}{3\left(y_1^2+y_2^2+y_3^2\right)} + \frac{3y_1^2}{3\left(y_1^2+y_2^2+y_3^2\right)} = \frac{y_3^2-2y_2y_3+y_2^2+3y_1^2}{3\left(y_1^2+y_2^2+y_3^2\right)}$$

solution x=y=z=2

Model:2

If number of variables are not equal to number of equations then that system may or maynot have solution.If at all solution exists then problem definetly cannot be solved directly.Some strategy is required to solve them That is not unique for all the problems.Here I discuss some of its kind, by going through it definetly I can say that your level of problem solving capability increases.

Let us have a glance at such problems

5. Find the positive integer solution of the equation $x+\dfrac{1}{y+\dfrac{1}{z}}=\dfrac{10}{7}$.

Sol: Given $x+\dfrac{1}{y+\dfrac{1}{z}}=1+\dfrac{3}{7}$

If number of variables and number of equaitons are same then equation is solvable. Above problem is an equation contains three unkown positive integers.

$y+\dfrac{1}{z}>1$ since $y,z\in N \Rightarrow 0<y+\dfrac{1}{z}<1$

x is positive integer $\Rightarrow x = 1$; $\dfrac{1}{y+\dfrac{1}{z}}=\dfrac{3}{7}$

$\Rightarrow y+\dfrac{1}{z}=\dfrac{7}{3}=2+\dfrac{1}{3} \Rightarrow y = 2$; $z = 3$

Therefore $x = 1$; $y = 2$; $z = 3$

6. Find all solutions in positive integers of the equation $\dfrac{1}{x}+\dfrac{1}{y}+\dfrac{1}{z}=\dfrac{7}{15}$.

Sol: Let x, y, z be positive integers such that $x\le y\le z$ then

17. If $P(x) = x^{15} - 2004x^{14} + 2004x^{13} - ..... - 2004x^2 + 2004x$ **.Then find *P*(2003) ?**

Sol:

Given $P(x) = x^{15} - 2004x^{14} + 2004x^{13} - ..... - 2004x^2 + 2004x$

$= x^{14}(x-2003) - x^{13}(x-2003) + x^{12}(x-2003) + ... - x(x-2003) + x$

$\therefore P(2003) = 0 - 0 + 0 + ... - 0 + 2003 = 2003$

18. In a triangle $h_a = 12$**,** $h_b = 20$ **.Prove that** $7.5 < h_c < 30$ **?**

Sol:

In any triangle difference of two sides is less than the third side.

$|a-b| < c$, $\Delta = \frac{1}{2}ah_a = \frac{1}{2}bh_b = \frac{1}{2}ch_c$

$\Rightarrow \left|\frac{2\Delta}{h_a} - \frac{2\Delta}{h_b}\right| < \frac{2\Delta}{h_c} \Rightarrow \left|\frac{1}{h_a} - \frac{1}{h_b}\right| < \frac{1}{h_c}$

$\Rightarrow \left|\frac{1}{12} - \frac{1}{20}\right| < \frac{1}{h_c} \Rightarrow \left|\frac{2}{60}\right| < \frac{1}{h_c} \Rightarrow \frac{1}{h_c} > \frac{1}{30} \Rightarrow h_c < 30$

$a + b < c \Rightarrow \frac{2\Delta}{h_a} + \frac{2\Delta}{h_b} < \frac{2\Delta}{h_c} \Rightarrow \frac{1}{h_a} + \frac{1}{h_b} < \frac{1}{h_c}$

$\Rightarrow \frac{1}{12} + \frac{1}{20} < \frac{1}{h_c} \Rightarrow \frac{8}{60} < \frac{1}{h_c}$

$\Rightarrow h_c > 7.5 \quad \therefore 7.5 < h_c < 30$

19. The polynomial $4x^4 - ax^3 + bx^2 - cx + 5 = 0$ has four positive real roots such that $\frac{r_1}{2} + \frac{r_2}{4} + \frac{r_3}{5} + \frac{r_4}{8} = 1$ **,find them ?**

Sol :

By Vieta, we have $r_1\ r_2\ r_3\ r_4 = \frac{5}{4}$

On the other hand, by AM-GM.

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A -2B +3C =14 $\Rightarrow$ (A,B,C) = (1 ,-2 ,3) ,(a ,b , c) = (2, -1 , 4)

10. Find x + y where x and y are nonzero solutions of the system of equations $y^2x = 15x^2 + 17xy + 15y^2$,$x^2y = 20x^2 + 3y^2$?

Sol :

Let $\frac{x}{y}$ = t Hence,

$$\frac{x^2y}{y^2x} = \frac{20x^2+3y^2}{15x^2+17xy+15y^2x} \Rightarrow t = \frac{20t^2+3}{15t^2+17t+15}$$

$\Rightarrow 5t^3 - t^2 + 5t - 1 = 0$

$\Rightarrow t = \frac{1}{5}$. Thus, y = 5x and x = 19 , y = 95.

11. Determine $x^2 + y^2 + z^2 + w^2$ If

$$\frac{x^2}{2^2-1} + \frac{y^2}{2^2-3^2} + \frac{z^2}{2^2-5^2} + \frac{w^2}{2^2-7^2} = 1$$

$$\frac{x^2}{4^2-1} + \frac{y^2}{4^2-3^2} + \frac{z^2}{4^2-5^2} + \frac{w^2}{4^2-7^2} = 1$$

$$\frac{x^2}{6^2-1} + \frac{y^2}{6^2-3^2} + \frac{z^2}{6^2-5^2} + \frac{w^2}{6^2-7^2} = 1$$

$$\frac{x^2}{8^2-1} + \frac{y^2}{8^2-3^2} + \frac{z^2}{8^2-5^2} + \frac{w^2}{8^2-7^2} = 1 \text{ ?}$$

Sol :

First let α = 4, 16, 36, and 64. Then our equation becomes

$$\frac{x^2}{\alpha-1} + \frac{y^2}{\alpha-9} + \frac{z^2}{\alpha-25} + \frac{w^2}{\alpha-49} = 1 \quad . \quad \text{--------} \quad (1)$$

Now, multiply (1) by $(\alpha-1)(\alpha-9)(\alpha-25)(\alpha-49)$ on both sides.

Then we would get a fourth degree polynomial in terms of α.

Also the roots of this polynomial are 4 , 16 , 36 , 64 or $(\alpha-4)(\alpha-16)(\alpha-36)(\alpha-64) = 0$.

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TRICKY PROBLEMS

1. 1002004008016032 has a prime factor p > 250000, find it?

Sol:

Let a = 10^3, b = 2

1002004008016032 = $a^5 + a^4b + a^3b^2 + a^2b^3 + ab^4 + b^5$

$= \frac{a^6 - b^6}{a-b} = (a + b)(a^2 + ab + b^2)(a^2 - ab + b^2)$

= 4.4.1002.250501.249501

Therefore a factor p > 250000 is 250501

2. Find only solution in R for

$(4x^2 + 6x + 4)(4y^2 - 12y + 25) = 28$?

Sol:

$A = (4x^2 + 6x + 4)(4y^2 - 12y + 25)$

$= ((2x + \frac{3}{2})^2 + \frac{7}{4})((2y - 3)^2 + 16)$

If $(x,y) \neq (\frac{-3}{4}, \frac{3}{2})$: $A > 16.\frac{7}{4} = 28$

then A = 28 is possible only when $(x, y) = \left(\frac{-3}{4}, \frac{3}{2}\right)$

3. Given that x,y,z are positive real numbers satisfying xyz =32, find the minimum value of $x^2 + 4xy + 4y^2 + 2z^2$.

Sol :

$E = x^2 + 4xy + 4y^2 + 2z^2 = x^2 + 2xy + 2xy + 4y^2 + z^2 + z^2$

Hence $\frac{E}{6} = \frac{x^2 + 2xy + 2xy + 4y^2 + z^2 + z^2}{6} \geq \sqrt[6]{x^2.2xy.2xy.4y^2.z^2.z^2}$

$= \sqrt[6]{16(xyz)^4} = \sqrt[6]{2^{24}} = 16$

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